RIPHAH INTERNATIONAL UNIVERSITY

Final Year Project: Household Relocation

Project Team

Student ID	Student Name	Program	E-mail
42895	Hafiz Muhammad	03321448139	42895@students.riphah.edu.pk
42912	Syed Raza Hussain	03338878205	42912@students.riphah.edu.pk

Supervisor(s) Name	Designation	Affiliation	E-mail
Mr. Asim Mansha	Senior lecturer		

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Introduction

- Moving to a new home is often stressful and hard to manage.
- Our platform brings everything into one place to make moving easier.
- It allows:
- Customers to book, pay, and track moves
- Movers to manage tasks
- Admins to oversee the whole process
- Truck owner's register their trucks
- Key features:
- GPS tracking
- Alerts and notifications
- Email Notifications
- Digital invoices and reviews
- The goal is to make relocation smooth, fast, and reliable for all.

Problem Statement

Home relocation is often stressful and overwhelming.
Customers face issues like:
•Unreliable moving services
•Confusing or inconsistent pricing
•No real-time move updates
•Poor communication
•Difficulty scheduling logistics
Truck owners struggle with:
•Disorganized task assignments
•No centralized booking system
•Miscommunication with clients and admins
Overall, the process is time-consuming, chaotic, and uncertain for all parties.

Solution

1. **All-in-One Platform**
   Book moving services with real-time tracking.
2. **Secure Login & Data Protection**
   Keeps user data safe with strong security.
3. **Smart Booking & Task Handling**
   Automatically connects customers with movers and warehouse staff.
4. **Customer Dashboard**
   View payments, get alerts, and manage your profile easily.
5. **Admin Panel**
   Control users, services, schedules, and payments in one place.
6. **Tools for Movers & Managers**
   Get tasks, update progress, and manage storage space smoothly.
7. **Safe Payments**
   Pay securely using trusted gateways.
8. **Live Tracking & Delivery Proof**
   Track services in real-time with digital confirmation.
9. **Fast & Easy Experience**
   Saves time, reduces hassle, and improves service for everyone.

Features and Scope

Feature Number	Feature Name	Description	Priority
1	User Authentication & Security	Secure registration, login, role-based access, password recovery.	High
2	Service Booking & Scheduling	Book relocation/storage, schedule via calendar, track reference numbers.	High
3	Task Assignment & Management	Admin assigns tasks, movers receive notifications, status updates.	High
4	Payment Processing & Billing	Secure payments, multiple methods, invoicing, refunds, and late fees.	High
5	Notifications & Communication	Email/SMS notifications, in-app messaging, feedback collection.	High
6	Dashboard & Reporting	Dashboards for admins, customers, and staff with service & financial data.	High
7	Service Tracking & Real-Time Updates	GPS tracking, proof of delivery with digital signatures.	High
8	Reviews	Customers can Review the services,	Medium
9	Distance-Based Pricing	Calculates the relocation cost based on the distance between the pickup and destination locations.	High

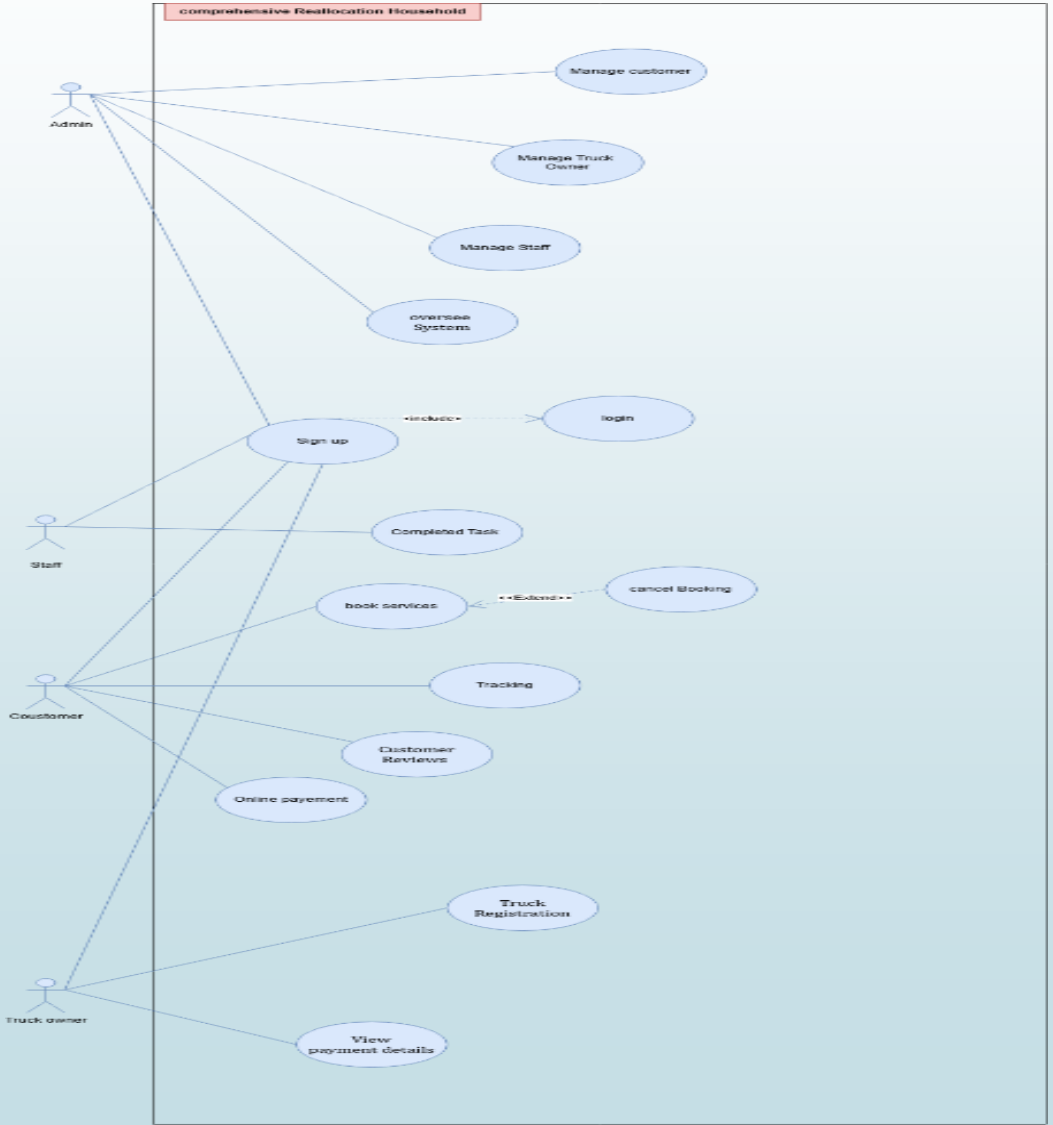
Requirements

- List of Different Users
  - Admin
  - Truck Owner
  - Customer
  - etc.
- Use Cases
- Functional & Non Functional Requirements

List of Different Users

User Class	Characteristics & Privileges
Customer	General public users; can book moves, make payments, receive updates, and give reviews.
Admin	Internal platform managers; control over bookings, finances, user accounts, and reporting.
Movers/Staff	On-ground personnel; receive task schedules and provide live updates.
Truck Owners	Fleet partners; register trucks, receive bookings, and manage payment records.

Use Case Diagram



Functional Requirements

- 1. Booking and Scheduling Service
- 2. Real-Time Move Tracking
- 3. Secure Payments and Invoicing
- 4. Role-Based Access Control (RBAC)

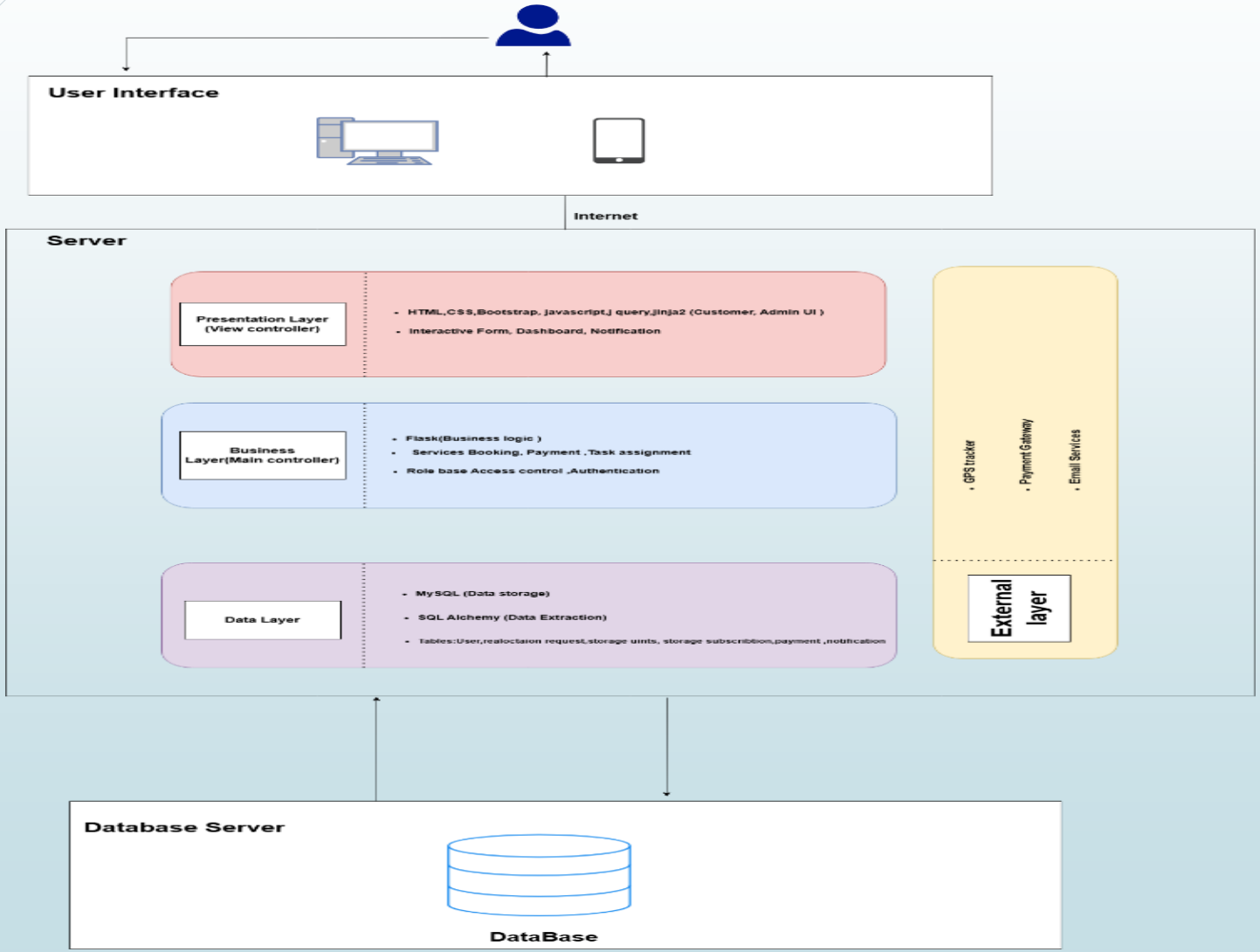
Non Functional Requirements

- 1. Performance Requirements
- 2. Safety Requirements
- 3. Security Requirements
- 4. Software Quality Attributes
- 5 Business Rules

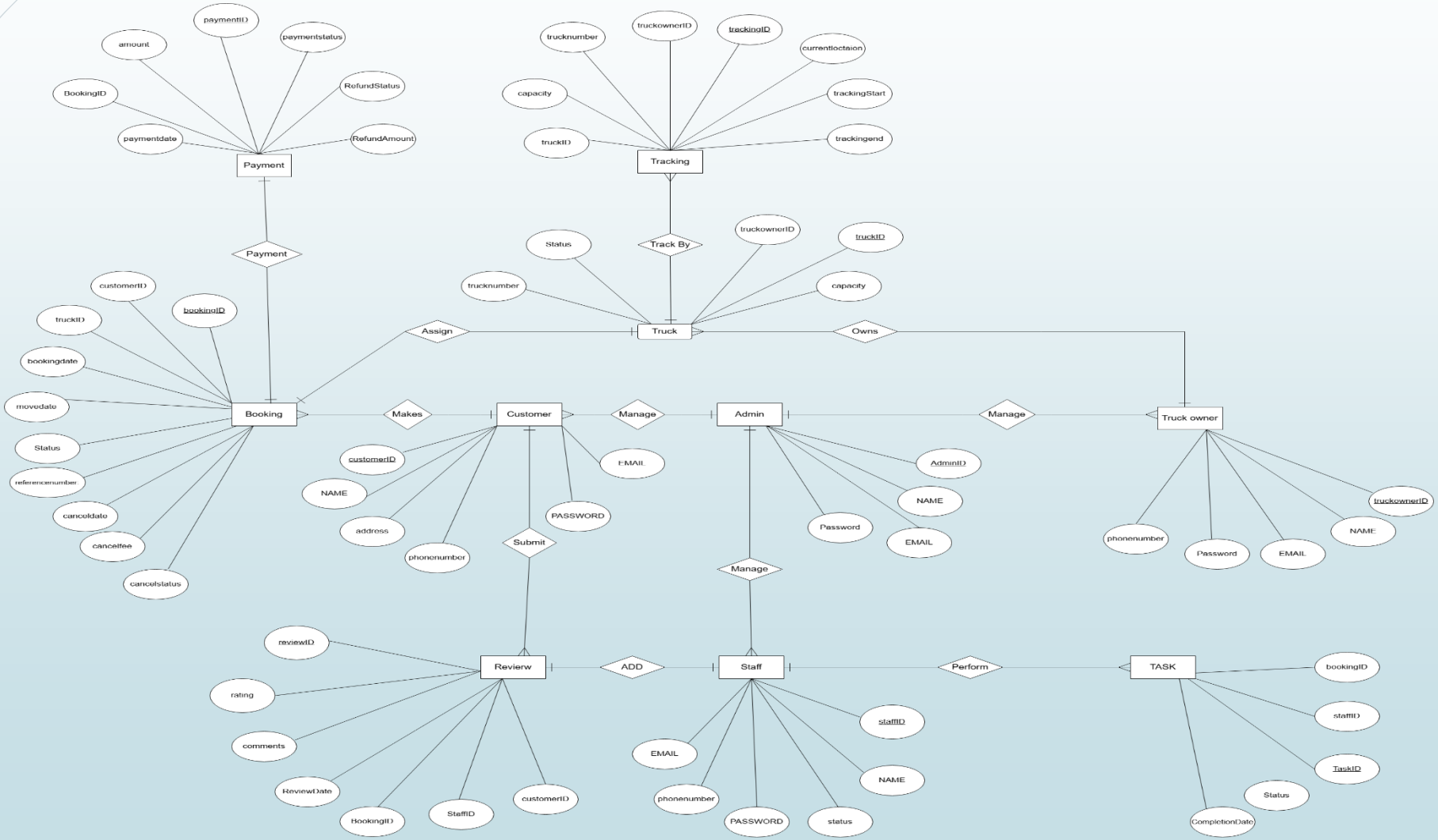
System Design

- Architecture Design
- Entity Relationship Diagram
- Class Diagram
- Sequence / Collaboration Diagram

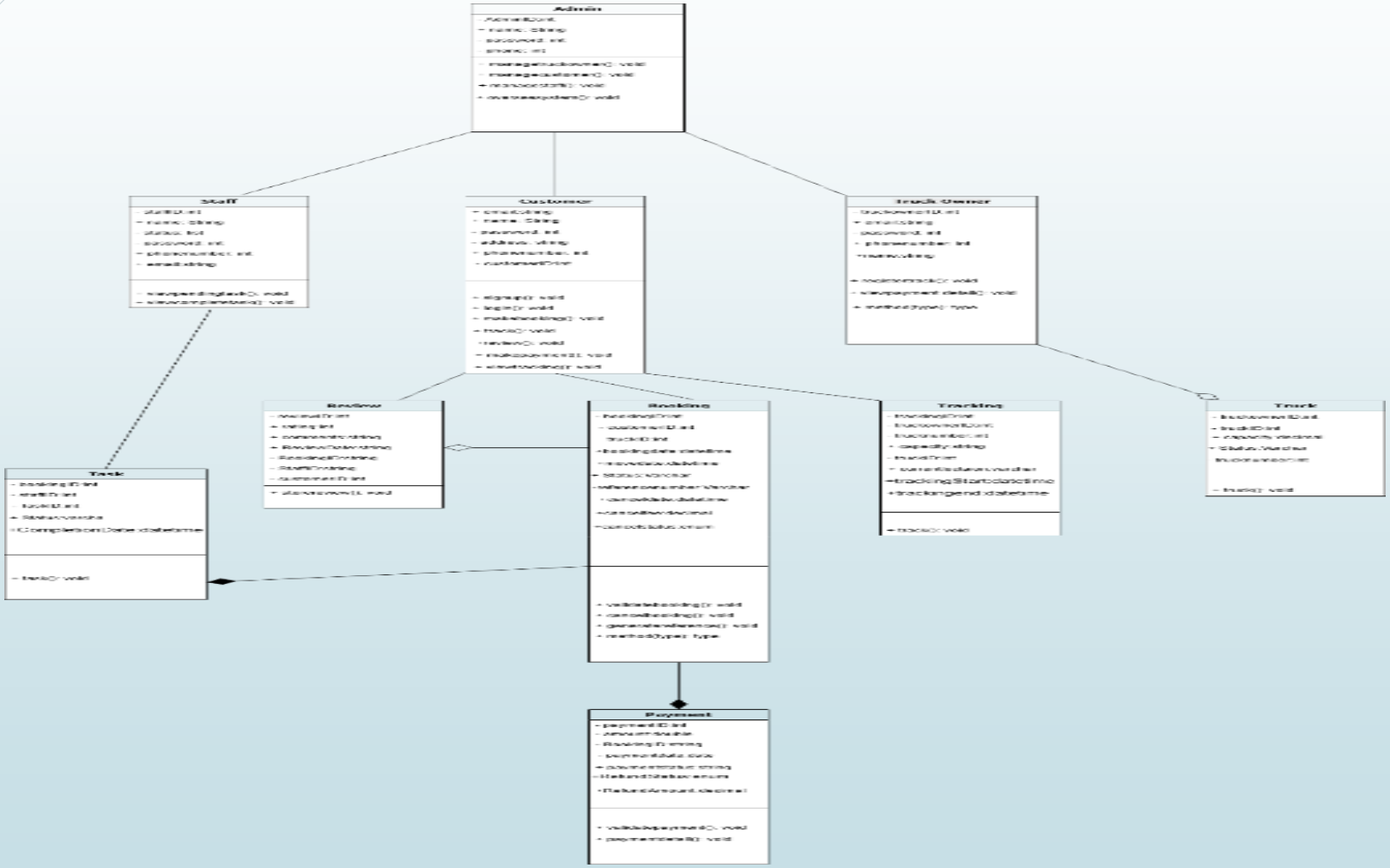
Architecture Design



Entity Relationship Diagram

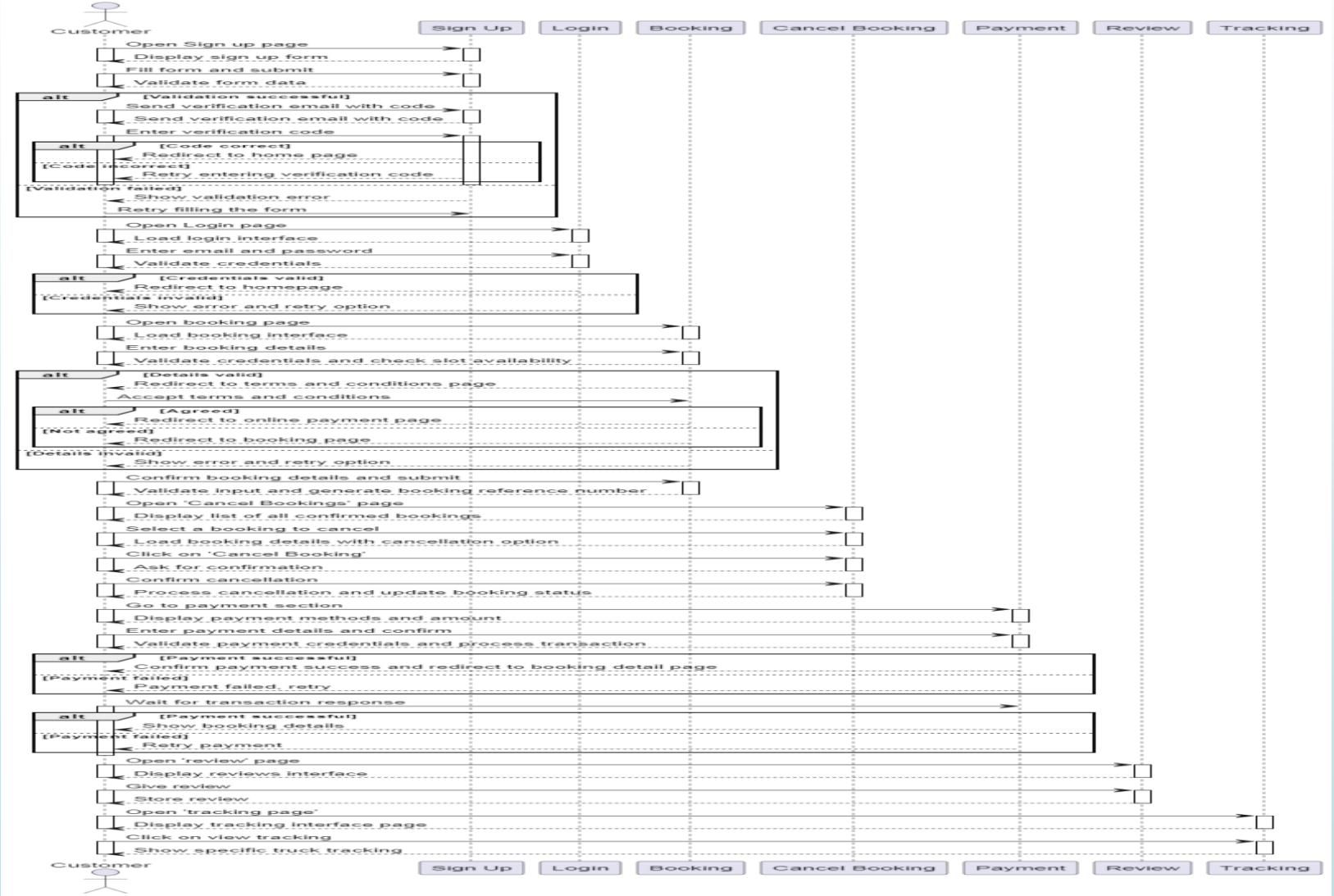


Class Diagram



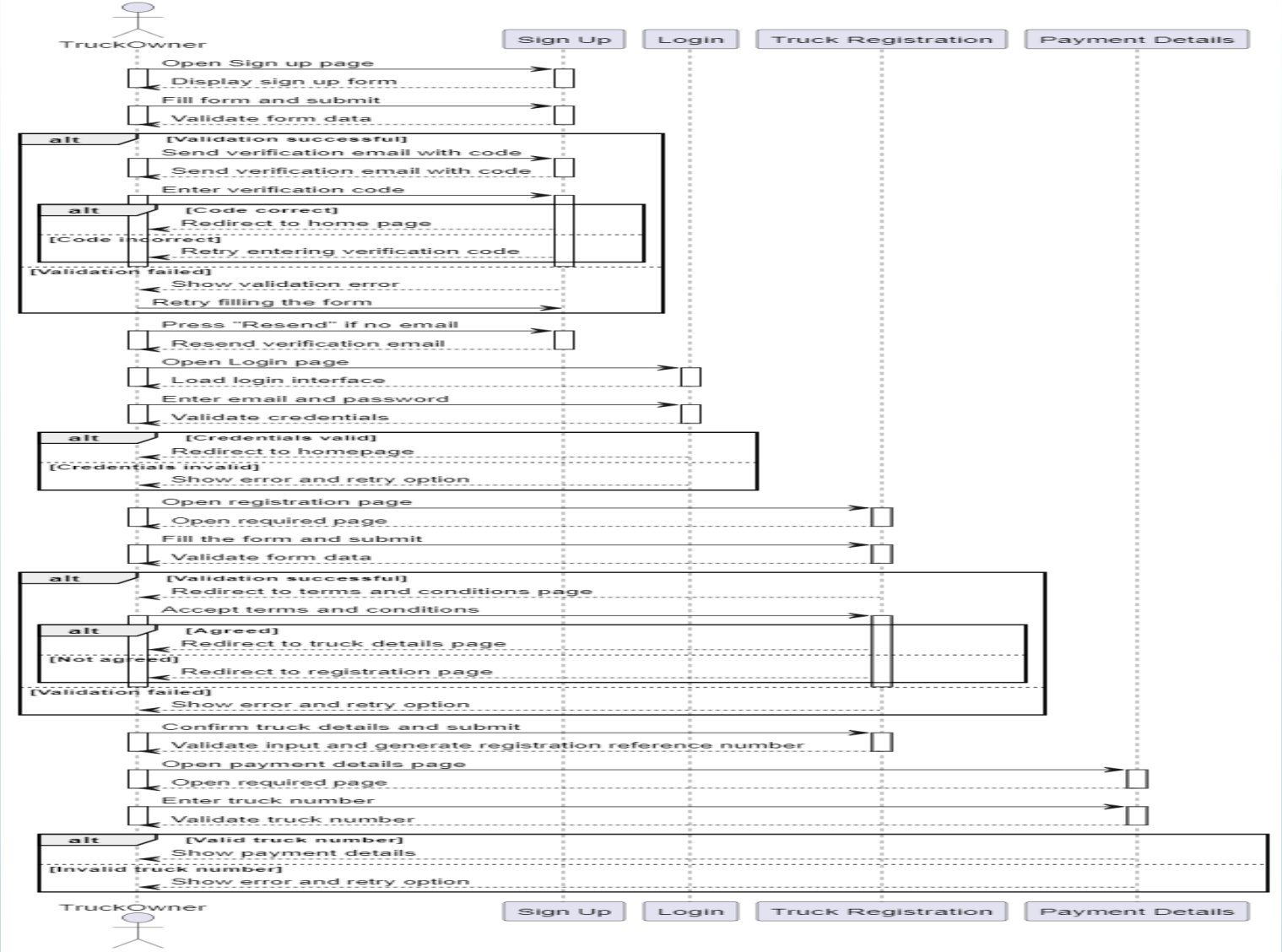
Sequence / Collaboration Diagram

- **Customer:**



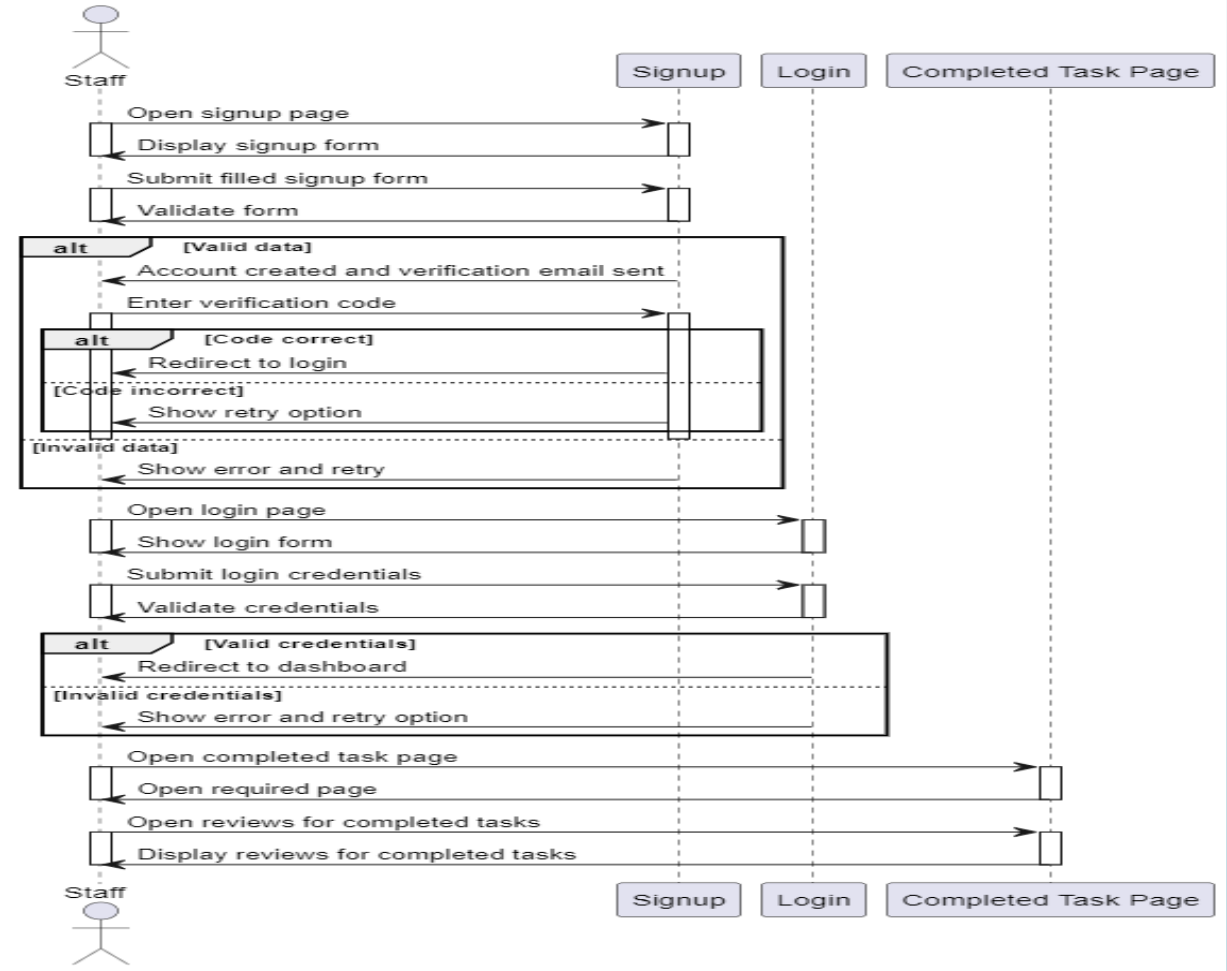
Sequence / Collaboration Diagram

- **Truck Owner:**



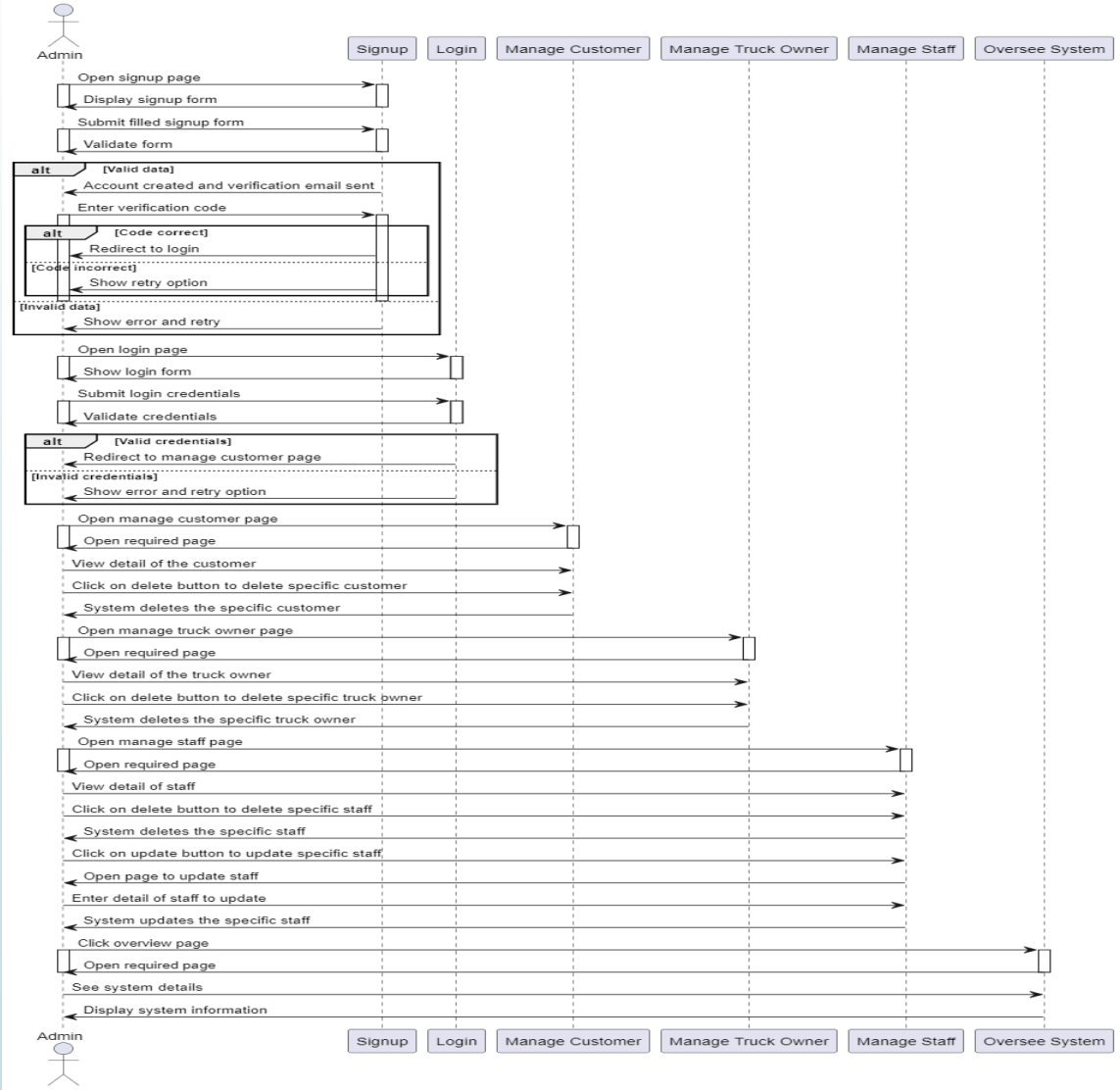
Sequence / Collaboration Diagram

- **Staff:**



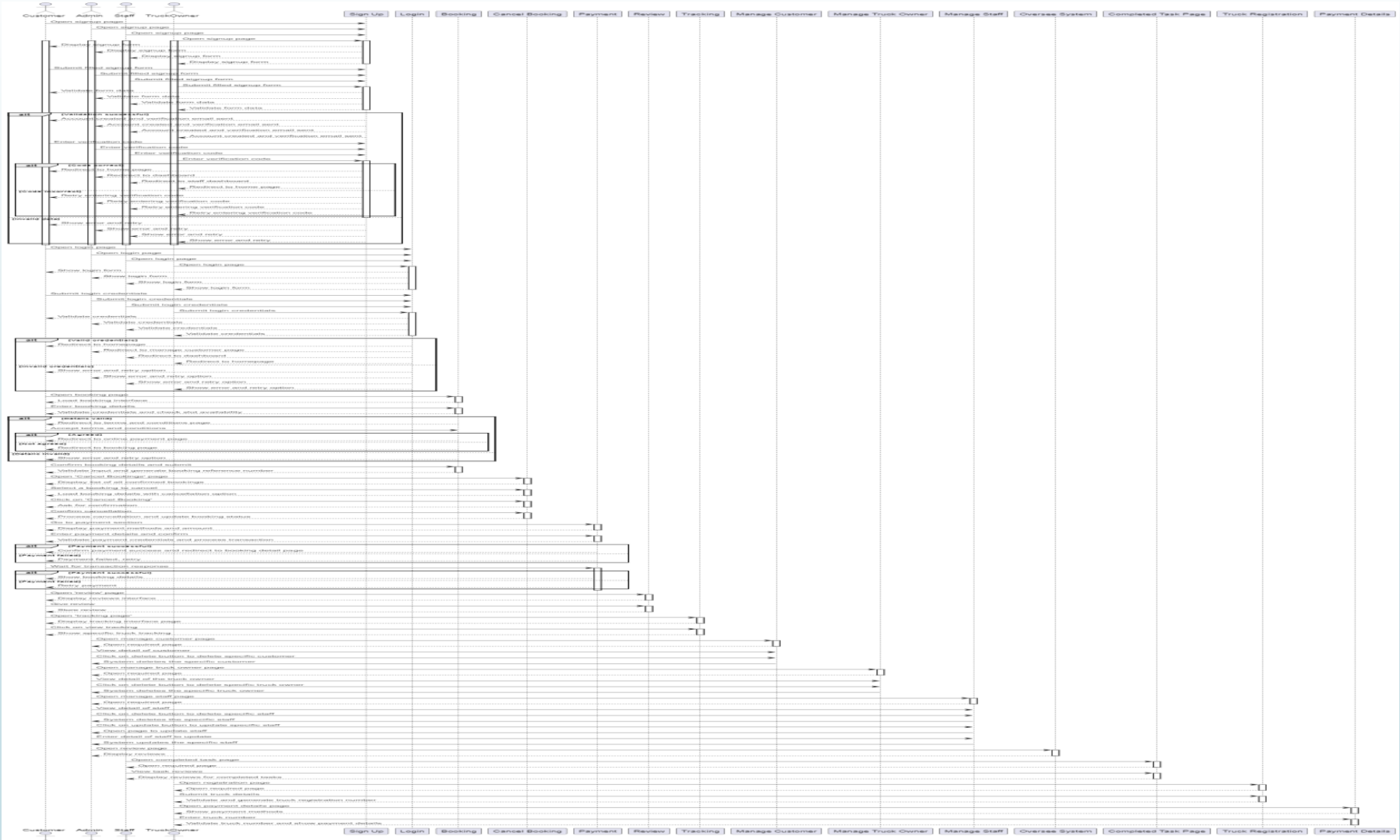
Sequence / Collaboration Diagram

- **Admin:**



Sequence / Collaboration Diagram

- **Overall:**



Implementation

- **Development Tools & Technologies:**

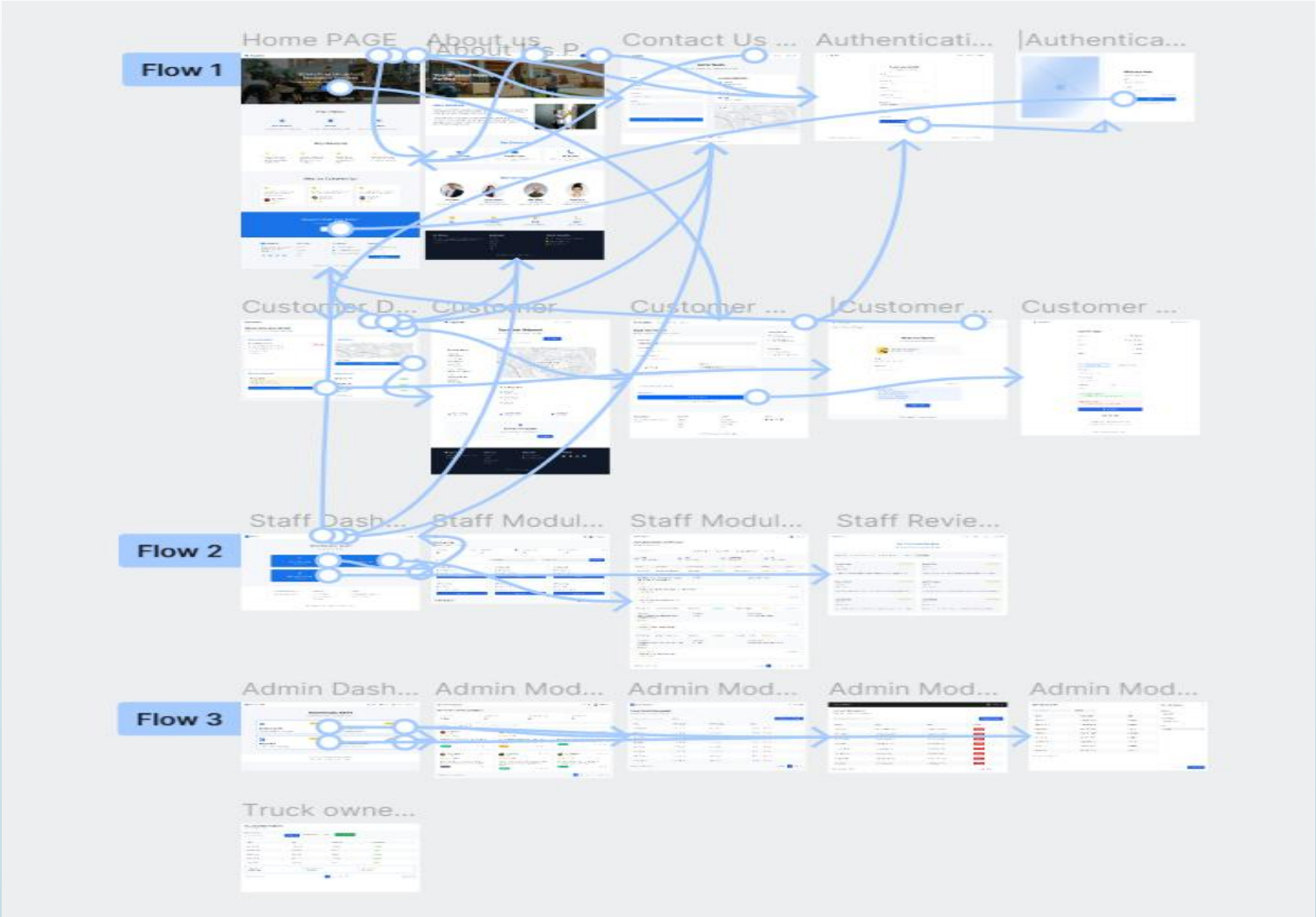
  The system architecture will consist of the following components:

- **Frontend**: A responsive web interface built with HTML, CSS, Bootstrap, JavaScript JQuery, and Jinja2 to provide an interactive and easy-to-use platform.
- **Backend**: The backend, developed in python flask, will handle service requests, user authentication, and payment processing.
- **Database**: A MySQL database and SQL Alchemy (ORM model) will store user data, provider listings, booking records, and transaction history.

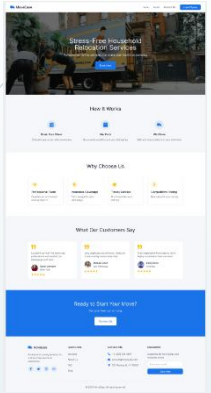
Challenges

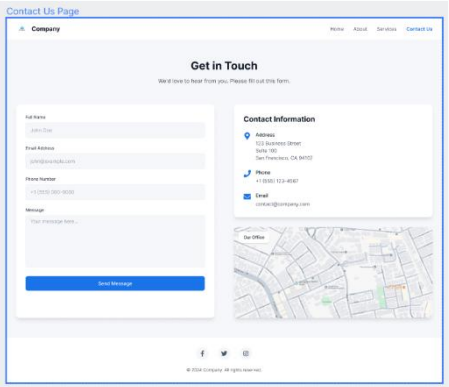
- Ensuring real-time GPS tracking accuracy and availability across regions
- Integrating secure online payments with fraud prevention
- Designing an intuitive UI for diverse users (customers, movers, admins)
- Managing concurrent tasks and booking schedules across roles
- Ensuring data privacy and compliance with user information policies
- Handling edge cases (e.g., cancelations, delayed moves, storage unavailability)

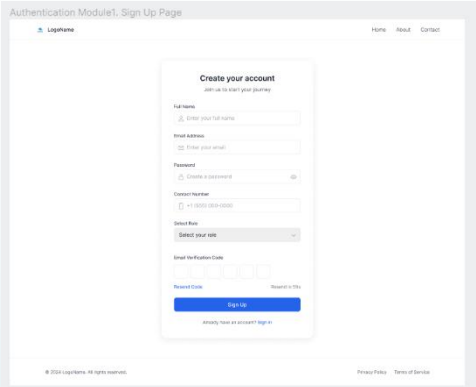
Prototype

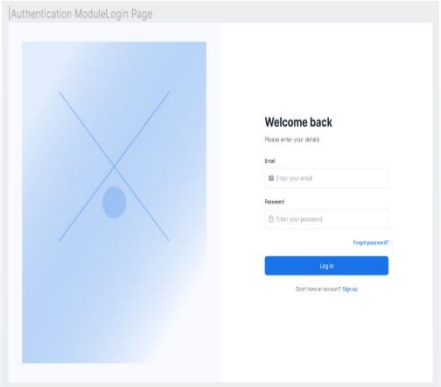

Flow 1
Home PAGE
About us
Contact Us ...
Authenticati...
Authentica...
Customer D...
Customer ...
Customer ...
Customer ...
Flow 2
Staff Dash...
Staff Modul...
Staff Modul...
Staff Revie...
Flow 3
Admin Dash...
Admin Mod...
Admin Mod...
Admin Mod...
Admin Mod...
Truck owne...

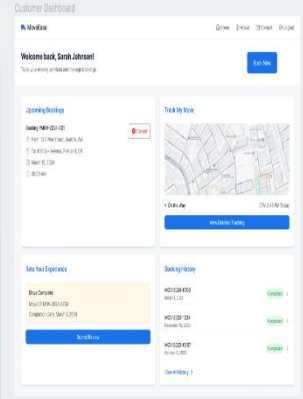
Screenshots

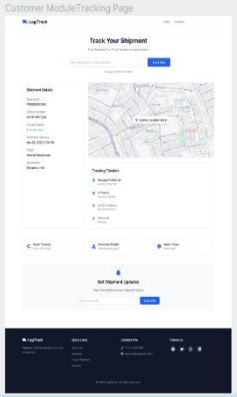


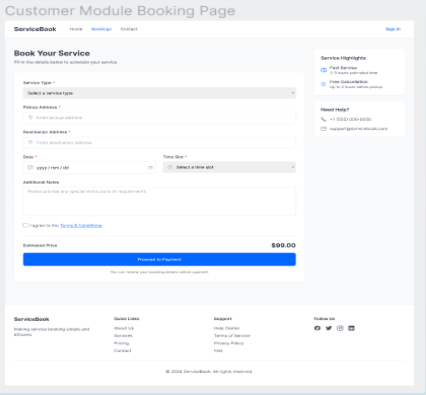


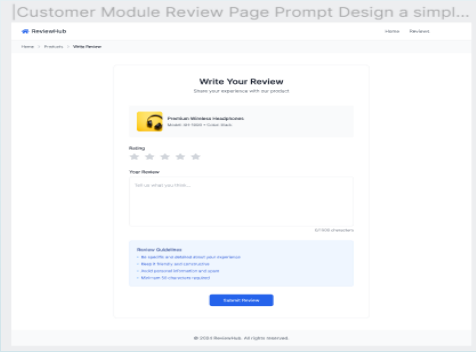


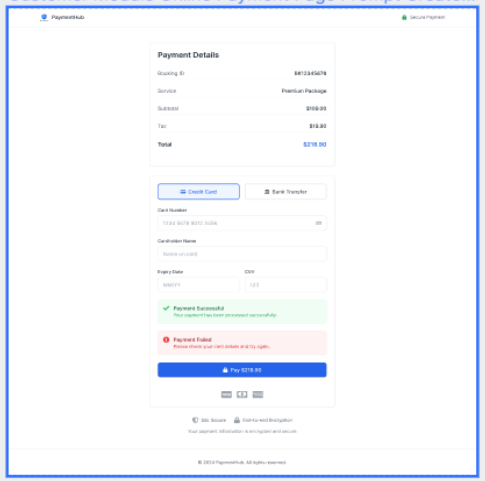






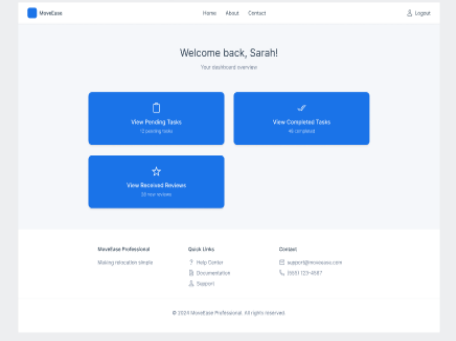




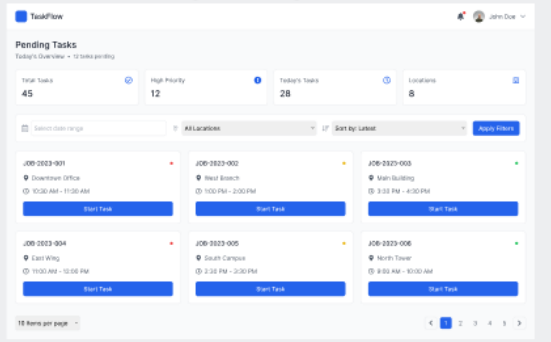


Screenshots

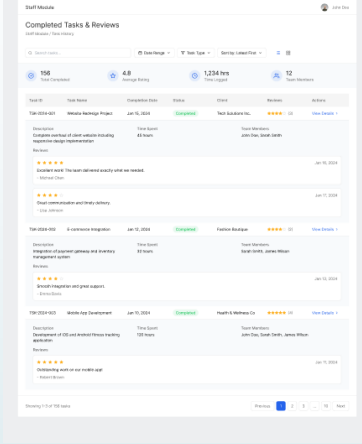
Staff Dashboard page



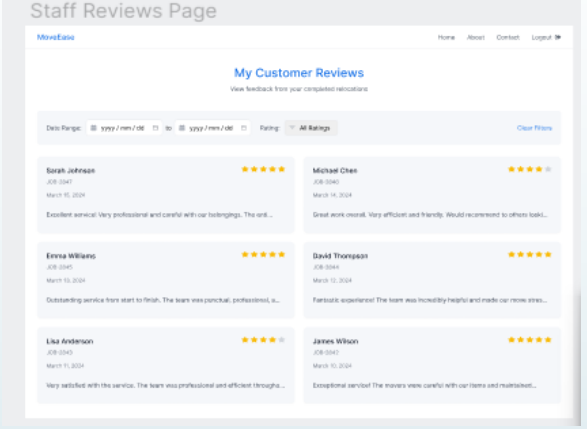
Staff ModulePending Tasks Page



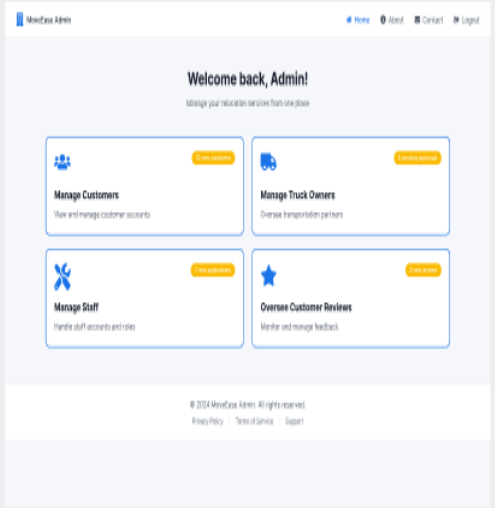
Staff ModuleCompleted Tasks & Reviews Page



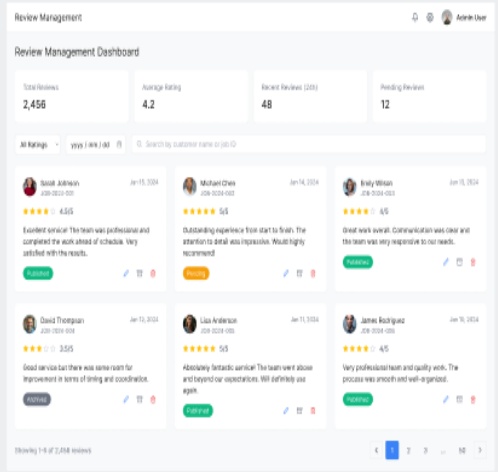
Staff Reviews Page



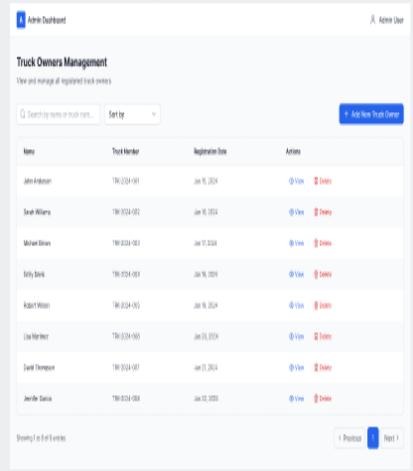
Admin Dashboard Page



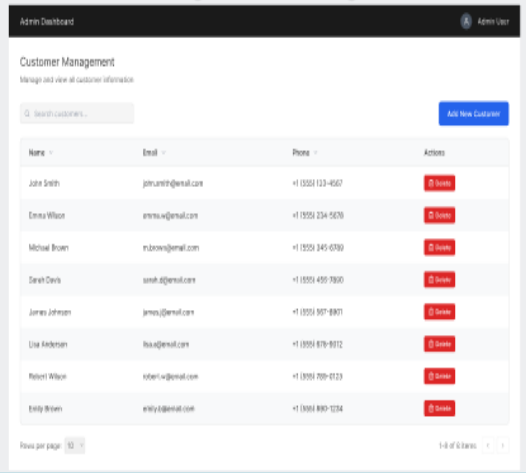
Admin ModuleOversee Reviews Page



Admin ModuleManage Truck Owners Page



Admin Module Manage Customers Page



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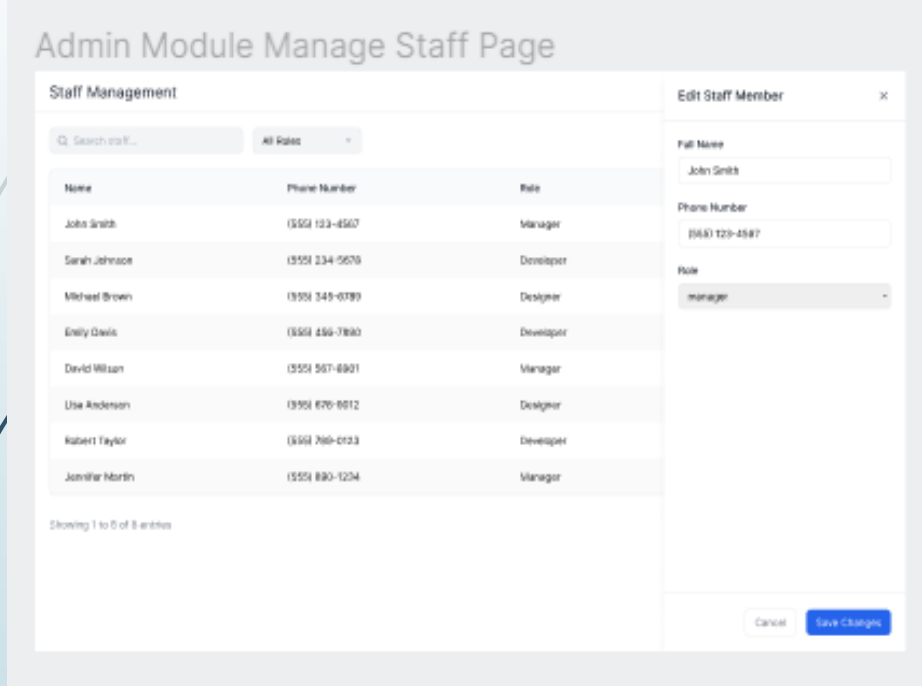

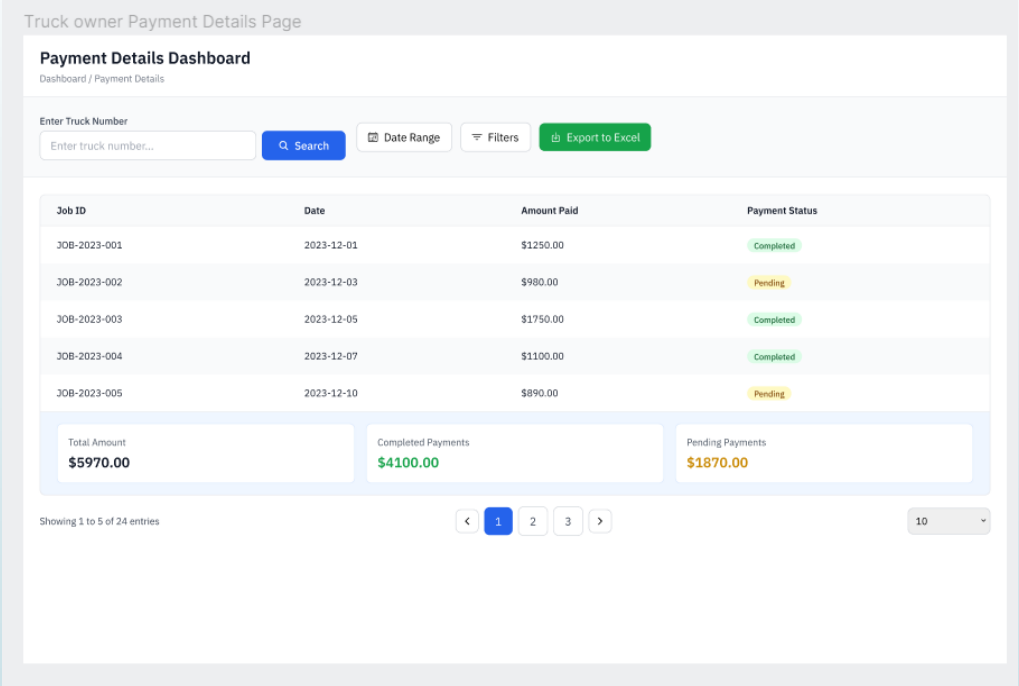

Report

- Link to report: Final_Project_Report_PPT.pptx